

Experimental Evaluation of a Multi-Layer Abstraction Approach for RL on Non-Markovian Tasks

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Outline

- ▶ Motivation
- ▶ Framework and Implementation
- ▶ Experimental Evaluation



Motivation and Problem Statement

1 Motivation

Sparse Rewards

Informative feedback
is received only after
reaching the final goal.

Non-Markovian Task

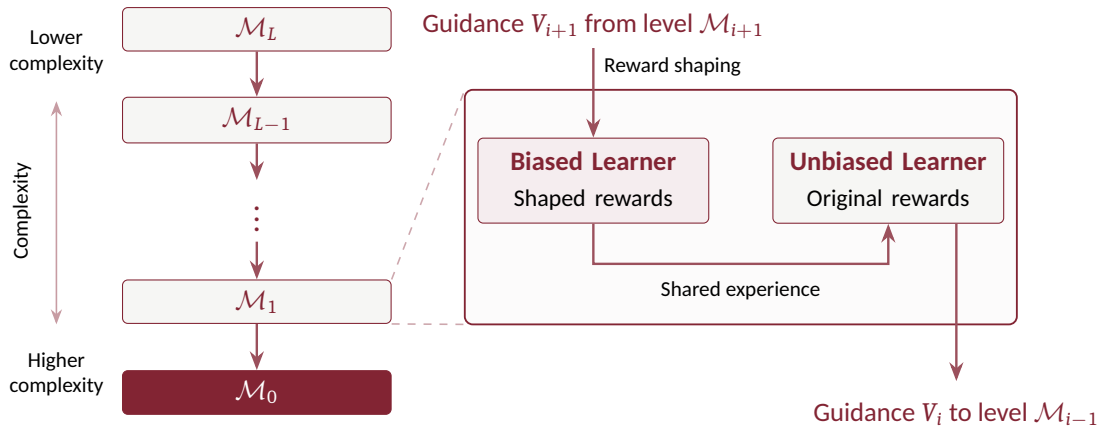
Task completion may depend
on the history of previously
observed events.

Prohibitive Learning



The Starting Point

2 Framework and Implementation

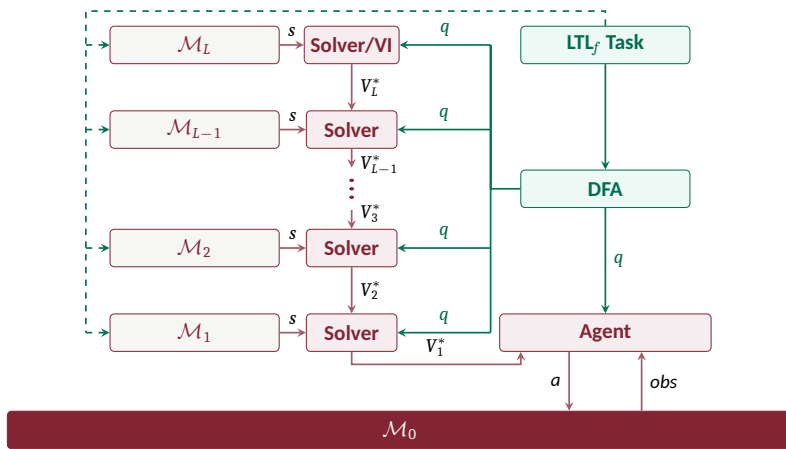


Framework by Cipollone et al. [1]



Extension Proposed in This Thesis

2 Framework and Implementation



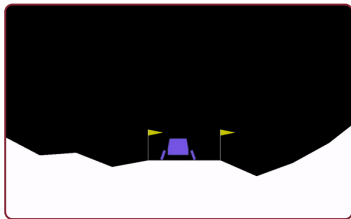


Experimental Setting

3 Experimental Evaluation

Ground Domain M_0

Gymnasium Lunar Lander



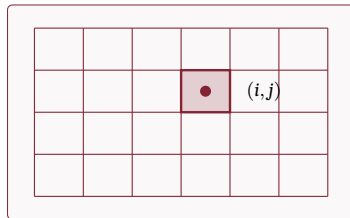
- $S = (x, y, v_x, v_y, \theta, \omega, c_l, c_r)$
- $\mathcal{A} = \{\text{no-op, eng-left, eng-right, eng-main}\}$
- Original dense reward removed

State mapping



Abstract Domain M_i

Grid Environment



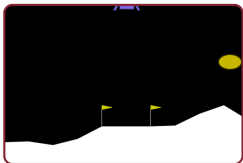
- $\mathcal{S}_i = \{(i,j) : 1 \leq i \leq H, 1 \leq j \leq W\}$
- $\mathcal{A} = \{\uparrow, \nearrow, \rightarrow, \searrow, \downarrow, \swarrow, \leftarrow, \nwarrow\}$
- Physical quantities discarded



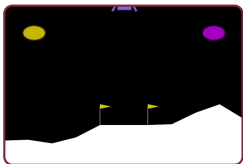
Biased-Unbiased Interaction: Practical Limitation

3 Experimental Evaluation

Required Tasks

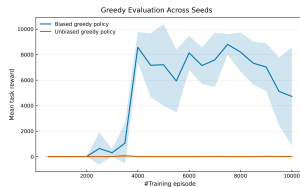
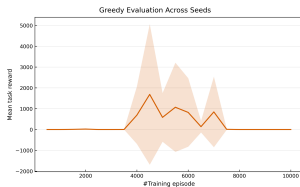
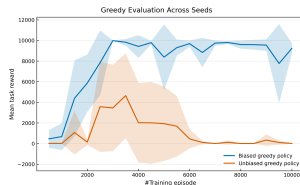
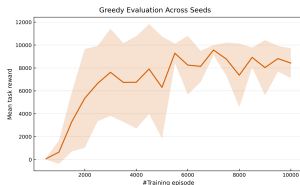


1-phase: $\diamond g_1$



2-phase: $\diamond(wp_1 \wedge \bigcirc \diamond g_1)$

Training results



Greedy eval.: standalone unbiased (left), coupled biased/unbiased (right); phases 1/2 (top/bottom).



Reward Shaping Under Temporal Misalignment

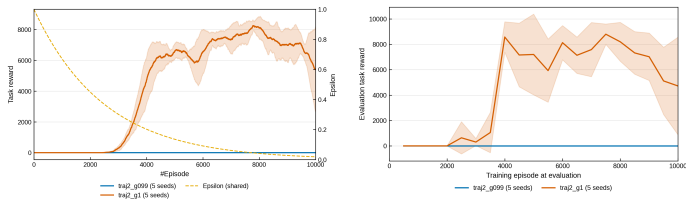
3 Experimental Evaluation

Shaping Formulation

$$\begin{cases} F(s, s') = \gamma_s \Phi(s') - \Phi(s) \\ \Phi(s) = V(s) \end{cases}$$

Potential Based Reward Shaping (PBRs) [1, 2].

Training Results



γ_s	Best success rate (%)	Same-state fraction (%)	Same-state reward per transition	Same-state reward per episode
0.99	0.00 $\pm$ 0.00	82.39 $\pm$ 0.96	-89.51 $\pm$ 0.05	-5000.60 $\pm$ 53.21
1.00	96.94 $\pm$ 1.91	91.48 $\pm$ 0.22	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00

Training reward (left) and greedy evaluation (right), with transition statistics for $\gamma_s = 0.99$ and $\gamma_s = 1$.



From Theory to Practice

3 Experimental Evaluation

Theory

Coupled architecture

Biased + Unbiased

PBRS Formulation

$$F(s, s') = \gamma_s V(s') - V(s)$$

Practice

Biased Learner

Only biased on the ground MDP

Shaping Heuristic

$$F(s, s') = V(s') - V(s)$$

Interaction
Problem

Temporal
Mismatch



Thank you for listening!
Any questions?



References

3 Experimental Evaluation

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